Bicycle anti-theft security systems

"What bicycle anti-theft locking systems have been developed between 2021 and 2026, and what technologies do they use?What are the limitations of existing smart bicycle locks that use RFID, Bluetooth, IoT, or electronic authentication? How effective are existing bicycle security systems at preventing unauthorized access and bicycle theft? Understand existing solutions and identify what they already accomplish." *Generated by LeapSpace*

Bicycle Anti-Theft Locking Systems (2021–2026): Quick Insights

Key Findings Table

Topic	What existing systems do	Main technologies	Main limitations	Evidence
Smartphone-controlled smart locks	Enable app-based unlocking, status checks, and remote monitoring	Bluetooth/BLE, smartphone apps, cloud/web interfaces, sometimes NFC	Security depends heavily on pairing design, phone trust, and app/cloud security; weak state handling and replay-style issues remain documented	[1] [2] [3] [4] [5] [6] [7] [8]
Sensor-rich bicycle anti-theft systems	Detect tampering or unauthorized movement and notify owners	Accelerometers, gyroscopes, vibration sensors, GPS, Wi-Fi/GSM/internet alerts	Good for alerting and deterrence, but evidence for stopping determined theft is limited	[9] [10] [11] [12] [13] [14] [15]
IoT-connected tracking and monitoring systems	Provide live location, remote visibility, and recovery support	ESP32, GPS, web dashboards, Telegram/internet notifications, nearby-phone/iBeacon reporting	Often omit offline-mode, link-security, and cloud-hardening details; broaden attack surface	[9] [16] [17] [18] [19] [20]
Electronic authentication beyond keys	Add barriers to unauthorized access	Fingerprint, PIN/keypad, OTP, NFC, multi-factor combinations	Convenience-security trade-offs; extra access paths and slower unlock workflows can weaken assurance or usability	[21] [22] [23] [24] [25] [26] [27]
RFID in bicycle-specific locks	Sparse bicycle-specific evidence	RFID appears mainly in generic access control papers	No clear bicycle-specific 2021–2026 evidence on enrollment, key storage, and unlock logic	[2] [10] [28] [29] [30] [31] [32]
Reported effectiveness	Best-supported outcomes are alerts, tracking, and recovery assistance	Monitoring, sensing, notification, GPS/location systems	Little rigorous evidence on false accepts/rejects, adversarial bypass, latency, battery cost, or broad real-world theft reduction	[4] [9] [15] [29] [33] [34] [35]

Direct Answer

Between 2021 and 2026, the retrieved bicycle anti-theft systems are mostly **hybrid smart-lock or monitoring systems** rather than purely mechanical locks: they combine app control, Bluetooth/BLE or Bluetooth-adjacent links, GPS tracking, accelerometer/gyroscope/vibration sensing, web or mobile alerts, and sometimes PIN, OTP, NFC, or biometric access control [1] [9] [10] [12] [15] [21].

What they already accomplish best is **owner notification, tamper detection, remote monitoring, and post-theft recovery support**. Some systems also add remote unlocking and access control for authorized users, and alarm-based deterrence can discourage opportunistic theft [9] [13] [14] [15] [29] [33].

Their main limitations are not a lack of features but **security assurance gaps**: insecure BLE pairing, replay/relay/spoofing exposure, excessive trust in the smartphone app, weak cloud/mobile/web interfaces, weak firmware/update security, poor secret storage, and limited reporting on adversarial testing [5] [19] [36] [37] [38].

For **RFID specifically**, the supplied evidence does **not** show a well-documented bicycle-specific RFID smart lock from 2021–2026 with clear tag enrollment, stored credentials, and unlock workflow; RFID evidence is mostly from generic access-control systems instead [2] [10] [28] [29].

Overall, current systems appear **more effective at detection and recovery than at proven prevention of determined theft**. Evidence for strong unauthorized-access prevention or real-world theft-rate reduction remains limited and uneven [4] [15] [34] [35].

Confidence level: High - multiple sources converge on the same pattern: feature-rich connected systems are common, but robust security validation is sparse [1] [9] [19] [34].

Study Scope

This report covers evidence retrieved for **2021–2026** on bicycle anti-theft locking systems and closely related two-wheeler or smart-lock security literature, with emphasis on **authentication methods, connectivity, sensing, vulnerabilities, and reported effectiveness** [9] [18] [34]. It includes bicycle-specific prototypes, monitoring systems, bike-sharing security work, and adjacent smart-lock/IoT analyses used to interpret lock attack surfaces [4] [7] [34] [36].

Assumptions & Limitations

- The evidence base is stronger for **prototypes, concepts, and adjacent IoT/smart-lock studies** than for a complete catalog of commercial bicycle lock products [4] [15] [39].
- Bicycle-specific RFID evidence is sparse, so RFID conclusions rely partly on the absence of documented bicycle implementations in the retrieved set [10] [29].
- Many studies report **functionality or architecture**, not standardized security metrics such as FAR/FRR, latency, battery impact, or adversarial bypass performance [4] [36] [40].

Suggested Further Research

- Standardized adversarial testing of bicycle smart locks against **BLE relay, replay, spoofing, app compromise, and credential-sharing attacks** [6] [7] [8].
- Bicycle-specific studies combining **digital attack resistance and mechanical forced-entry resistance** in one evaluation [4] [39].

- Long-term field deployments measuring **real theft reduction, unauthorized-access incidents, recovery rates, and maintenance burden** [15] [34].
- Better reporting of **latency, battery drain, false rejects, false accepts, and offline behavior** across authentication methods [4] [40].
- Bicycle-specific implementations of **secure RFID/NFC credential management** with documented enrollment and key storage [2] [28] [29].

Full Report Body

Introduction

From 2021 to 2026, bicycle anti-theft development trends shifted toward **connected, sensor-rich, electronically authenticated systems** rather than standalone locks [1] [9] [15]. Across the retrieved literature, the common goal is not only to block unauthorized access, but also to detect tampering, notify the owner quickly, and improve the chance of recovery after theft [9] [29] [33].

Taken together, these findings suggest that “bicycle anti-theft locking systems” in this period should be understood as a spectrum: from app-controlled locks, to sensor-equipped alarm locks, to IoT tracking-and-alerting platforms, to broader bike-sharing security systems [1] [10] [17] [34].

Confidence level: High - this framing is directly supported across multiple source clusters [1] [9] [15].

Foundational Context

A useful way to evaluate these systems is to separate three security functions:

1. **Access control**: who is allowed to unlock or operate the bicycle [21] [22] [24].
2. **Tamper/theft detection**: whether the system can notice movement, vibration, displacement, or misuse [10] [11] [13].
3. **Recovery support**: whether it helps locate or manage the bicycle after theft [9] [41].

This distinction matters because many systems perform well on detection and recovery but provide only limited evidence of strong lockout security against determined attackers [4] [9] [34]. A GPS tracker may help recover a bike without preventing initial theft, while a BLE lock may offer convenient access control yet still be exposed to protocol attacks if pairing is weak [6] [8].

A second foundational point is that security depends not only on the visible lock hardware but on the **full trust chain**: phone app, pairing protocol, cloud interface, firmware, update mechanism, and storage of secrets [5] [19] [38].

Confidence level: High - the distinction between access prevention, alerting, and recovery recurs throughout the evidence [9] [15] [29] [33].

Technologies Used in Bicycle Anti-Theft Systems

Bluetooth/BLE and App-Controlled Unlocking

App-controlled unlocking is one of the most common patterns in the retrieved material [1] [2] [3]. One bicycle smart-lock concept combined a smartphone/Android app with NFC, GPS, Bluetooth, and accelerometer-based authentication, although the pairing or handshake details were not specified [4]. Another connected-lock study paired the lock to a smartphone over Bluetooth with cloud-linked access control, but authorization state residing on the phone created state-consistency risk [5].

More general BLE security analyses show that protection against replay, spoofing, and man-in-the-middle threats depends heavily on the pairing mode. Weak modes such as **Just Works** lack strong authentication, while stronger options include authenticated secure connections, passkey, numeric comparison, out-of-band methods, and certificate-oriented extensions [42] [43] [44] [45].

So, BLE is widely useful for convenience and remote control, but its security value depends on implementation quality rather than on BLE use alone [7] [36] [46].

IoT Connectivity, GPS, and Remote Monitoring

Several bicycle and two-wheeler systems use IoT connectivity to provide tracking, location visibility, and notifications [9] [16] [18]. A 2025 smart bike security system used **ESP32 + GPS** with a web interface for remote unlocking/monitoring and real-time location updates [9]. A 2024 bicycle monitoring/finding framework used IoT sensors plus nearby smartphones and iBeacon-like reporting for stolen-bike location detection [17]. A 2023 monitoring system provided internet-connected alerts on a web page [18].

These designs clearly expand visibility and recovery support, but summaries often omit cloud dependence, offline behavior, and communication-security details [9] [17] [18]. That omission matters because IoT literature repeatedly identifies compromised cloud interfaces, weak authentication, absent encryption, and undersecured gateways as recurring attack surfaces [19] [20] [37] [47].

Sensor-Assisted Theft Detection

Another strong trend is adding sensing to detect suspicious events [9] [10] [11] [12]. Sensors used include accelerometers, gyroscopes, vibration sensors, GPS, and in one broader vehicle context even alcohol sensors [9] [10] [11] [12] [48].

In bike-oriented implementations, LockX combined an electronic lock with accelerometer and vibration sensing plus immediate Wi-Fi alerts [15]. An anti-theft electric-bike design used vibration and positioning sensors with network notifications, so displacement, vibration, or battery removal could trigger owner alerts [14]. Another vibration-sensor system reported accurate theft-attempt detection and low-cost protection, emphasizing alarm/notification-based deterrence [13].

These systems are strongest when the goal is early warning and deterrence, not necessarily cryptographic prevention of access [13] [14] [15].

PIN, Biometrics, OTP, NFC, and Multi-Factor Methods

The literature also includes electronic authentication beyond phone proximity [21] [22] [23] [24] [25]. Multi-factor smart locks combine RFID, keypad/password, app control, OTP, or biometrics to strengthen authorization [21] [22] [23] [24] [25].

A fingerprint-based two-wheeler system replaced a physical ignition/key interface with biometric access and sent GSM alerts on mismatch, while also adding a 4-digit keypad for family access [26]. A Wi-Fi bike lock used both static PIN and rotating local PIN authentication; the rotating PIN increased unlock time from about 6-8 seconds to 9-13 seconds, trading speed for stronger access control [27].

Taken together, these findings suggest a trend toward layered authentication, but the evidence base says much more about architecture than about comparative security metrics [21] [22] [24] [40].

RFID

RFID appears in the broader access-control literature, and one learning notes RFID-based bicycle/vehicle locks can provide passive authentication and Wi-Fi/app-linked alerts [15]. However, the focused bicycle-specific retrieval did **not** identify a bicycle anti-theft prototype or commercial bicycle lock with documented RFID tag enrollment, credential storage, and unlock authorization workflow [10] [29]. The RFID papers found were mostly generic access-control systems that authorize access after reading a registered tag [2] [28] [30] [31] [32].

So the retrieved evidence supports RFID as a general access-control technique, but not as a well-documented bicycle-specific smart-lock implementation in this 2021–2026 corpus [2] [29].

Confidence level: High - bicycle-specific RFID evidence is consistently described as sparse or absent in the retrieved set [10] [29].

Limitations and Attack Surfaces

BLE and Smartphone Trust Weaknesses

BLE-related weaknesses are among the clearest documented problems [6] [7] [36]. Reported attack vectors include compromised shareable credentials, SDK exploitation, replay/time-reset attacks, message tampering, and pairing weaknesses [6] [8] [36]. Relay attacks also remain relevant in proximity-based BLE systems [49].

One broader study found that many commercial BLE devices remained vulnerable despite claimed protections, and a 2026 device study reported that fewer than 5% used secure pairing, with some bypassing standardized pairing entirely [7] [36]. Although these were not all bicycle locks, they are directly relevant because bike locks rely on the same security primitives [7] [46].

A particularly important limitation is **state inconsistency** when authorization lives primarily on the smartphone instead of the lock itself [5]. This means convenience features can undermine assurance if the phone or app becomes the weak link [5] [8].

IoT Cloud, Mobile, and Firmware Risks

IoT connectivity broadens attack surface substantially [19] [37] [47] [50] [51]. Recurring problems include weak authentication, poor encryption, insecure cloud/web/mobile interfaces, weak device management, and vulnerable gateways [19] [20] [37] [47].

Firmware/update weaknesses are repeatedly highlighted: insecure or missing update mechanisms, lack of automatic updates, interrupted OTA updates, and exposure to malicious firmware during delivery [38] [52] [53] [54] [55] [56]. Even secure boot can be bypassed through hardware or software vulnerabilities [57] [58]. Secure update pipelines still depend on safe certificate storage, trusted execution, and uncompromised servers or companion apps [59] [60].

For bicycle systems specifically, many summaries do not explain how offline operation, transport security, or trusted key storage are handled [9] [18]. That is a practical security gap because bike locks are often battery-powered and intermittently connected [61].

Electronic Authentication Tradeoffs

Electronic authentication adds barriers, but often with tradeoffs [26] [27]. The fingerprint-plus-keypad design adds convenience for family or relatives, but that also creates an additional access path [26]. The rotating-PIN design improves protection but increases unlock delay [27].

This pattern implies a recurring design tension: more convenient access flows can increase exposure, while stronger controls can increase friction [5] [26] [27].

RFID-Specific Limitations

Because bicycle-specific RFID implementations are not well documented in the retrieved set, their limitations cannot be compared in a mature bicycle-specific way [10] [29]. What is documented is only the generic model: a registered RFID tag is read and authorization follows [2] [28]. That is not enough to assess bicycle-specific risks such as tag enrollment security, key revocation, stolen-tag handling, or passive-keyless relay resistance [2] [31] [32].

Confidence level: High - BLE and IoT attack surfaces are well supported; RFID-specific bicycle evidence remains thin [19] [29] [36].

Effectiveness at Preventing Unauthorized Access and Theft

What existing systems clearly accomplish

The strongest evidence supports four capabilities:

1. **Authentication barriers for authorized use** through app control, PIN, biometrics, or multi-factor logic [15] [21] [22] [24] [62].
2. **Tamper/motion detection** via accelerometer, gyroscope, or vibration sensors [10] [11] [13] [15].
3. **Immediate owner alerts** through Wi-Fi, GSM, internet, or Telegram-style messaging [1] [9] [14] [16].
4. **Recovery support** through GPS/location tracking and web/mobile visibility [9] [41].

These capabilities are meaningful. They can deter opportunistic theft, notify the owner quickly, and improve the chance of locating a stolen bicycle [9] [29] [33].

What is less clearly proven

Evidence is much weaker for **measured resistance to unauthorized access by determined attackers** [4] [36]. The retrieved bicycle papers do not report false accept/false reject rates, standard authentication latency comparisons, battery impact, or consistent adversarial bypass testing [4] [10] [15] [40] [62]. Controlled attack evaluations against bicycle devices are largely absent [15] [35].

A bike-sharing study reported simulations and field testing suggesting reduced theft rates using an SVM+IoT security approach, but it also noted challenges in real-time monitoring, low-latency IoT connectivity, and data preparation [34]. A 2025 rental-lock system claimed scalability in controlled environments, but did not provide strong real-world theft-reduction validation [15].

So the current literature supports functionality and plausibility more strongly than verified, generalizable theft prevention [15] [34].

Prevention versus recovery

A major cross-source pattern is that many systems are **recovery-oriented rather than prevention-proven** [9] [41]. GPS tracking and crowd-assisted reporting help after loss [9] [17]. Vibration sensing and alarms help early detection and deterrence [13] [14]. But neither automatically proves that the lock itself can resist protocol abuse, forced entry, or advanced electronic bypass [4] [36] [39].

Confidence level: High - the literature consistently documents alerts and tracking more strongly than robust theft-prevention outcomes [9] [34].

Prototype Maturity and Validation Gaps

Many retrieved systems remain at the level of **concepts, prototypes, or controlled-environment demonstrations** [4] [15] [39]. Even where bicycle devices are implemented, summaries often omit detailed lock hardware, handshake schemes, or anti-tamper validation [4].

Validation methods are often indirect: scenario testing, simulations, field testing in limited contexts, spatial analysis of theft patterns, or user-perception studies [34] [35] [63] [64]. These methods can show feasibility or context, but they do not establish strong resistance to live adversaries [35] [36].

A major methodological weakness is the lack of standardized benchmarks for:

- false accepts and false rejects [4] [40]
- unlock latency and usability [4]
- battery and resource cost [4] [61]
- cloud/offline security behavior [9] [20]
- forced-entry plus digital-bypass testing together [4] [39]

Confidence level: High - the gap between security claims and evaluation quality is directly supported across the corpus [4] [35] [36].

Practical Synthesis: What Existing Solutions Already Accomplish

Current bicycle anti-theft systems already accomplish the following with reasonable consistency:

- **Authorized-user access control** through app-based, PIN-based, biometric, or layered electronic methods [21] [22] [24].
- **Tamper awareness** using motion, vibration, and displacement sensing [10] [11] [13].
- **Real-time owner notification** through internet, Wi-Fi, GSM, or cloud/app messaging [9] [15] [16].
- **Recovery assistance** through GPS or distributed reporting frameworks [9] [17] [41].
- **Operational visibility** via web dashboards or mobile monitoring [3] [9] [18].

But existing solutions only partially accomplish the harder goal of **provably resisting sophisticated unauthorized access and preventing theft under adversarial conditions** [36]. The main unresolved problems are weak BLE pairing, overtrust in phones and apps, cloud/API insecurity, firmware/update risk, and limited bicycle-specific security validation [5] [7] [19] [38].

Conclusion

From 2021 to 2026, bicycle anti-theft locking systems evolved toward **hybrid smart security platforms** combining app control, BLE/Bluetooth, GPS, sensors, and in some cases PIN, NFC, biometrics, or multi-factor logic [1] [9] [10] [15] [21]. The clearest evidence shows these systems already perform **monitoring, alerting, tamper detection, and recovery support** effectively in controlled or prototype settings [9] [15] [29] [33].

Their key weakness is that **security assurance lags behind feature growth**. Weak pairing, replay/relay exposure, insecure app/cloud interfaces, firmware/update weaknesses, and underreported validation leave many systems insufficiently proven against determined attackers [6] [19] [36] [38]. Bicycle-specific RFID evidence is especially sparse, and standardized comparisons across authentication methods are largely missing [29] [40].

Taken together, these findings suggest that existing smart bicycle locks are best understood today as **useful deterrence, detection, and recovery tools**, with only partial evidence that they can robustly prevent unauthorized access and theft across real-world adversarial conditions [15] [34].

Overall conclusion confidence: High - the evidence strongly supports the feature set and the validation gap, even though it does not support a complete product-by-product catalog or definitive theft-prevention rankings [1] [19] [34].

References

1. Jadhav, P., Gaul, S., Madhwai, A., (...), Deshmukh, M. (2023) A Cutting-Edge Security Solution: OTP-Based Smart Wireless Locking System. *2023 4th International Conference on Computation, Automation and Knowledge Management, ICCAKM 2023*. https://doi.org/10.1109/ICCAKM58659.2023.10449610
2. Harshitha, G., Gopika, S.N., Gourinandan, B., (...), Nalinakshan, S. (2025) RFID based Access Control System. *3rd International Conference on Electronics and Renewable Systems, ICEARS 2025 - Proceedings*. https://doi.org/10.1109/ICEARS64219.2025.10941511
3. Venkatesh, T., Karthikeyan, A. (2026) Secure Proximity-Aware Vehicle Access Using UWB and RF Command Control with BLE based Telemetry. *2026 International Conference on Next Generation Electronics, NELEX 2026*. https://doi.org/10.1109/NELeX69209.2026.11590579
4. Ali, Ahmad, Ahmed, Mansoor, Khan, Abid, (...), Helfert, Markus (2021) VisTAS: blockchain-based visible and trusted remote authentication system. *PeerJ Computer Science*. https://doi.org/10.7717/peerj-cs.516
5. Heiding, F., Süren, E., Olegard, J., Lagerström, R. (2023) Penetration testing of connected households. *Computers and Security*. https://doi.org/10.1016/j.cose.2022.103067
6. Feng, T., Wang, Y., Wu, Y. (2026) Workflow-level formal analysis of BLE pairing and a lightweight server-assisted key agreement framework. *Computers and Electrical Engineering*. https://doi.org/10.1016/j.compeleceng.2026.111094
7. Pöhls, H.C., Steffens, L., Völk, D. (2026) IoT security in the regulatory era: A case study on BLE pairing. *Computers and Security*. https://doi.org/10.1016/j.cose.2026.105106
8. Allen, A., Mylonas, A., Vidalis, S., Gritzalis, D. (2024) Smart homes under siege: Assessing the robustness of physical security against wireless network attacks. *Computers and Security*. https://doi.org/10.1016/j.cose.2023.103687
9. Sanjai Ragunath, M., Premalatha, H.M. (2025) Smart Bike Security System with Alcohol Detection and GPS Tracking. *Proceedings of the 2025 3rd International Conference on Inventive Computing and Informatics, ICICI 2025*. https://doi.org/10.1109/ICICI65870.2025.11069865

10. Cabalquinto, E.D.M., Malabayabas, L.M., Pulido, K.A.A., (...), Domingo Lopez, D.J. (2022) Anti-theft Integrated Bike Lock with Gyroscope Sensor and Accelerometer Motion Detection. *2022 IEEE 14th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management, HNICEM 2022*. https://doi.org/10.1109/HNICEM57413.2022.10109492

11. Mallieswaran, K., Agaramudhalvan, S., Nithya, R., (...), Radhika, S. (2024) Design and Implementation of Advanced Safety Features in Two-Wheeled Vehicles for Accident Prevention and Anti-Theft System. *SAE Technical Papers*. https://doi.org/10.4271/2024-01-5229

12. Goilkar, S.S., Prasad, S.V.G.V.A., Madhu, P., (...), Chandra, S. (2024) Biometric authentication and theft alert system for motorcycles using IoT. *Contemporary Solutions for Sustainable Transportation Practices*. https://doi.org/10.4018/979-8-3693-3755-4.ch015

13. Menaka, M., Tharani, R., Pemila, M., (...), Divyalakshmi, D. (2025) GSM Based Vehicle Anti-Theft Device Using Vibration Sensor And Arduino. *2025 International Conference on Data Science and Business Systems, ICDSBS 2025*. https://doi.org/10.1109/ICDSBS63635.2025.11031878

14. Huang, B.-C., Gao, Q.-H. (2022) Intelligent anti-theft system for electric bicycle based on Internet of Things. *Proceedings of SPIE - The International Society for Optical Engineering*. https://doi.org/10.1117/12.2637500

15. Vijay Sathappan, M., Sujatha, S., Rathish, A., (...), Tham, M.-L. (2025) Sustainable Smart Lock for Vehicle Rentals in Controlled Environment. *5th IEEE International Conference on Innovations in Power and Advanced Computing Technologies, i-PACT 2025*. https://doi.org/10.1109/i-PACT65952.2025.11307942

16. Yusro, M., Djatmiko, W., Kirana, C.A.P., (...), Pangestu, A. (2024) Mo-SSeS: A Motorcycle Smart Security System Using Raspberry Pi Based on the Internet of Things. *International Conference on Artificial Intelligence and Mechatronics System, AIMS 2024*. https://doi.org/10.1109/AIMS61812.2024.10512991

17. Chen, L.-W., Cho, C.-Y., Li, A.-N., Tsai, M.-F. (2024) Crowdsourced Bicycle Monitoring and Finding With Riding User Recognition Based on Internet of Things Technologies. *IEEE Sensors Journal*. https://doi.org/10.1109/JSEN.2024.3454233

18. Lata, S., Gowda, S.T., Spandana, K.P., (...), N, N. (2023) Smart Bike Monitoring System Using IoT. *Proceedings - 2023 7th International Conference on Design Innovation for 3 Cs Compute Communicate Control, ICDI3C 2023*. https://doi.org/10.1109/ICDI3C61568.2023.00032

19. Reshmi, T.R., Abhishek, K. (2024) 5G and 6G Security Issues and Countermeasures. *Secure Communication in Internet of Things: Emerging Technologies, Challenges, and Mitigation*. https://doi.org/10.1201/9781003477327-25

20. Slaibi, T., Ivaki, N., Vieira, M. (2026) IoT security assessment: A systematic literature review. *Journal of Systems and Software*. https://doi.org/10.1016/j.jss.2026.112833

21. Nayak, P., Gupta, G.P., Raj, V., (...), Sushma Shree, H.S. (2026) LockEase- Smart Door Security System. *Proceedings of 2nd International Conference on Visual Analytics and Data Visualization, ICVADV 2026*. https://doi.org/10.1109/ICVADV67766.2026.11470435

22. Tiwari, R., Arora, A., Srivastava, P.K., (...), Kumar, J. (2026) Multi-Layered Security System Utilizing RFID, Fingerprint, Keypad and IoT to Access the Lock. *Proceedings of the 2026 13th International Conference on Computing for Sustainable Global Development, INDIACom 2026*. https://doi.org/10.23919/INDIACom70271.2026.11526159

23. Vinoth Kumar, C.N.S., Vasim Babu, M., Naresh, R., (...), Bharathi, V. (2022) Real Time Door Security System With Three Point Authentication. *4th International Conference on Recent Trends in Computer Science and Technology, ICRTCST 2021 - Proceedings*. https://doi.org/10.1109/ICRTCST54752.2022.9782004

24. Thopate, K., Shinde, S., Mahajan, R., (...), Jadhav, S. (2023) Keyless Security The Smart Solution for Home with a Smart Door Lock. *International Journal on Recent and Innovation Trends in Computing and Communication*. https://doi.org/10.17762/ijritcc.v11i8s.7187

25. Singh, S., Gupta, A., Verma, J. (2025) Enhancing security with machine learning: A smart locking system approach. *Integrated Technologies in Electrical, Electronics and Biotechnology Engineering*. https://doi.org/10.1201/9781003606208-84

26. Dombale, A., Thakare, C., Thorve, M., (...), Zagade, S. (2024) Biometric Authentication System for Two Wheeler. *Proceedings - 2024 4th International Conference on Pervasive Computing and Social Networking, ICPCSN 2024*. https://doi.org/10.1109/ICPCSN62568.2024.00116

27. Painter, P., Waghmare, V., Katte, P., (...), Chavan, S. (2025) WiFi Lock for Bike Security with PIN Authentication. *International Conference on Emerging Technologies in Electronics and Green Energy, ICETEG 2025*. https://doi.org/10.1109/ICETEG66194.2025.11473153

28. Srikanth, A., Ellanti, P., Dakshitha, M.S., (...), Kalaiselvi, K. (2025) Secure-Tap: Enhancing Security with RFID-based Access Control. *16th International Conference on Advances in Computing, Control, and Telecommunication Technologies, ACT 2025*. https://www.scopus.com/pages/publications/105034842164

29. Paľačean, A.-V., Tranca, D.-C., Contasel, C., (...), Dutescu, C. (2021) IoT Enabled Optimized Architectures for GPS Anti-Theft Tracking Devices. *Proceedings - RoEduNet IEEE International Conference*. https://doi.org/10.1109/RoEduNet54112.2021.9638274

30. Kanagamalliga, S., Rajalingam, S., Deepasree, S., Jemi Chrisline, C. (2026) Revolutionizing Smart Home Security with an Arduino-Powered RFID Door Access Control System. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-3-032-06250-5_34

31. Goyal, V., Walia, A., Singh, M., (...), Dhaundiyal, R. (2023) Finger print based IoT for door lock system using RFID implementation over node MCU. *AIP Conference Proceedings*. https://doi.org/10.1063/5.0152783

32. el-Yakub, U.R., Jimpa, K., Mandal, A.K., Saha, H. (2023) Blynk, Database and WhatsApp Integrated RFID Based Lock System. *2023 26th International Conference on Computer and Information Technology, ICCIT 2023*. https://doi.org/10.1109/ICCIT60459.2023.10441308
33. Langga, M.N.H.A., Yuliana, M. (2024) Implementation of Twinguard: Motorcycle Theft Prevention Innovation System with Access Code and Shortest Route Recommendation. *Digest of Technical Papers - IEEE International Conference on Consumer Electronics*. https://doi.org/10.1109/ISCT62336.2024.10791292
34. Karthika, K., Nithya, A., Sujatha, M., (...), Murugan, S. (2024) Advancing Bike Sharing Security with SVM and IoT-Enabled Tracking and Anti-Theft Measures. *2nd International Conference on Sustainable Computing and Smart Systems, ICSCSS 2024 - Proceedings*. https://doi.org/10.1109/ICSCSS60660.2024.10625130
35. https://www.sciencedirect.com/science/article/pii/S1936645021002102
36. Lin, J., Chen, Z., Yang, X., (...), Jalil Piran, Md. (2025) Silent Intruders: Unveiling Security Flaws in BLE Smart Locks. *IEEE Transactions on Consumer Electronics*. https://doi.org/10.1109/TCE.2025.3619417
37. Thampan, M., Joy, R.T., Ramadani, L., Rajesh Kanna, R. (2025) Collaborative security approaches for IoT ecosystems. *IoT Security: Fundamentals and Key Enabling Technologies*. https://doi.org/10.1016/B978-0-443-34125-0.00026-X
38. https://www.sciencedirect.com/science/article/pii/S1573754320000512
39. Wen, D.-L., Huang, P., Qian, H.-Y., (...), Zhang, X.-S. (2021) Hybrid nanogenerator-based self-powered double-authentication microsystem for smart identification. *Nano Energy*. https://doi.org/10.1016/j.nanoen.2021.106100
40. Gupta, S., Kacimi, M., Crispo, B. (2022) STEP & TURN—Anovel bimodal behavioral biometric-based user verification scheme for physical access control. *Computers and Security*. https://doi.org/10.1016/j.cose.2022.102722
41. Gala, N., Poswalia, A., Gharat, R. (2023) Electric Bike Security: Biometric & GPS Integration for Intrusion Detection. *2nd International Conference on Sustainable Computing and Data Communication Systems, ICSCDS 2023 - Proceedings*. https://doi.org/10.1109/ICSCDS56580.2023.10105001
42. Varshney, C., Varshney, G. (2023) An improved authentication scheme for BLE devices with no I/O capabilities. *Computer Communications*. https://doi.org/10.1016/j.comcom.2023.01.001
43. https://www.sciencedirect.com/science/article/pii/S2730723925001411
44. Loubet, Gaël, Alata, Eric, Takacs, Alexandru, Dragomirescu, Daniela (2023) A Survey on the Security Challenges of Low-Power Wireless Communication Protocols for Communicating Concrete in Civil Engineerings. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s23041849
45. https://www.sciencedirect.com/science/article/pii/S2730723921000208

46. Sergiou, C., Iordanou, C., Katzis, K., (...), Didonet del Fabro, M. (2026) Security and risk assessment of IoMT communication protocols: A data-driven analysis of real-world vulnerabilities. *Internet of Things (The Netherlands)*. https://doi.org/10.1016/j.iot.2026.102085

47. Padmavathi, V., Saminathan, R. (2024) Security for the Internet of Things. *Computer and Information Security Handbook, Fourth Edition: Volumes 1-2*. https://doi.org/10.1016/B978-0-443-13223-0.00019-9

48. Balashanmuga Vadivu, P., Kaviya, S., Nandhitha, A., (...), Preethi, V. (2026) Embedded-based Smart Anti-Theft Backpack System for Real-Time Monitoring, Security, and Theft Prevention. *6th International Conference on Inventive Computation and Information Technologies, ICICIT 2026 - Proceedings*. https://doi.org/10.1109/ICICIT69063.2026.11634271

49. Gesteira-Miñarro, Roberto, López, Gregorio, Palacios, Rafael (2025) Revisiting Wireless Cyberattacks on Vehicles. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s25082605

50. Cambosuela, L., Kaur, M., Astya, R. (2024) The Vulnerabilities and Risks of Implementing Internet of Things (IoT) in Cyber Security. *2024 11th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions), ICRITO 2024*. https://doi.org/10.1109/ICRITO61523.2024.10522460

51. Ahmed, S.S., Abbasi, H., Ali, S.A., Owais, M. (2025) A Comprehensive Review of Security Challenges and Countermeasures in IoT Networks. *IMTIC 2025 - 8th International Multi-Topic ICT Conference: AI Driven Innovations Across Disciplines for a Resilient and Sustainable Infrastructure*. https://doi.org/10.1109/IMTIC68267.2025.11520530

52. Hashmat, F., Abbas, S.G., Hina, S., (...), Abbas, W. (2022) An automated context-aware IoT vulnerability assessment rule-set generator. *Computer Communications*. https://doi.org/10.1016/j.comcom.2022.01.022

53. Koulouras, Grigorios, Katsoulis, Stylianos, Zantalis, Fotios (2025) Evolution of Bluetooth Technology: BLE in the IoT Ecosystem. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s25040996

54. Charilaou, Christia, Lavdas, Spyros, Khalifeh, Ala, (...), Zinonos, Zinon (2021) Firmware Update Using Multiple Gateways in LoRaWAN Networks. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s21196488

55. Tariq, Muhammad Usman (2025) Challenges and solutions of Internet of Things security in the age of connectivity. *Human-Centric Integration of 6G-Enabled Technologies for Modern Society*. https://doi.org/10.1016/B978-0-443-27434-3.00022-2

56. Sey, C., She, S., Barnes, O., (...), Gyarteng, E.S.A. (2025) Blockchain-Enabled Adaptive Criticality-Based IoT Firmware Update Distribution Framework for Sustainable Smart Cities. *Journal of Industrial Information Integration*. https://doi.org/10.1016/j.jii.2025.100929

57. Reddy, R. Ravinder, Reddy, M. Venkata Krishna, Uyyala, Ravi, (...), Tyagi, Amit Kumar (2025) Wireless communications fundamentals and security requirements for future. *Human-Centric Integration of 6G-Enabled Technologies for Modern Society*. https://doi.org/10.1016/B978-0-443-27434-3.00009-X

58. https://www.sciencedirect.com/science/article/pii/S1573048424001900

59. https://www.sciencedirect.com/science/article/pii/S1936645021000061

60. Verderame, L., Ruggia, A., Merlo, A. (2023) PARIOT: Anti-repackaging for IoT firmware integrity. *Journal of Network and Computer Applications*. https://doi.org/10.1016/j.jnca.2023.103699

61. Rahmanikivi, M., Pérez-Solà, C., Garcia-Font, V. (2026) SmartBLock: a smart lock protocol over the Bitcoin blockchain. *Blockchain: Research and Applications*. https://doi.org/10.1016/j.bcra.2025.100350

62. Ismail, I.N., Zakariya, M.A.Z., Azman, N.S., (...), Hussin, M.S.F. (2021) Electronic development for a keyless anti-theft bicycle lock system prototype. *AIP Conference Proceedings*. https://doi.org/10.1063/5.0044264

63. Venverloo, Titus, Duarte, Fábio, Benson, Tom, (...), Ratti, Carlo (2023) Tracking stolen bikes in Amsterdam. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0279906

64. Márquez, L., Soto, J.J. (2021) Integrating perceptions of safety and bicycle theft risk in the analysis of cycling infrastructure preferences. *Transportation Research Part A: Policy and Practice*. https://doi.org/10.1016/j.tra.2021.06.017